

User Guide: ZIP Code Data Processor (ZipCode2.0)

Overview

This application processes ZIP code data from a CSV or structured file format. It allows users to:

- Convert standard CSV files into a structured, length-indicated format
 - Generate summaries of geographic extremes (east, west, north, south) by state
 - Search and display ZIP code records using command-line flags
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Launching the Application

Compilation

To compile the application:

```
g++ -o myProgram main.cpp CSVBuffer.cpp
```

Running

You can run the program in two modes:

1. Interactive Menu Mode (default)

```
./myProgram
```

This prompts the user to:

- Convert a CSV to a structured format
- Process a file to generate a summary

2. ZIP Code Search Mode

```
./myProgram ZipCodes -z56301 -z90210 -z99999
```

- `ZipCodes` is the file (CSV or structured)
 - Each `-z#####` is a ZIP code to search for
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File Formats

Input CSV

The CSV must have the following header:

```
"Zip Code","Place Name","State","County","Latitude","Longitude"
```

Length-Indicated Format

This format includes:

- Header metadata
- Records prefixed by their length

It is automatically generated using **option 1** in the interactive mode.

Features

Convert CSV to Structured Format

- Launch the program
- Choose option 1
- Provide input and output filenames

Generate State Summary

- Choose option 2
- Select a sort field: zipCode, placeName, state, latitude, or longitude
- Output is written to a CSV file (name provided by the user)

Lookup ZIP Codes (Command-line)

- Run with -z##### flags
 - Matching records are printed
 - If a ZIP code isn't found, a message is shown
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Sample Output

State Summary Output

```
State, Easternmost, Westernmost, Northernmost, Southernmost  
NY, 10001, 14905, 10598, 10002  
CA, 90210, 96162, 96162, 90001
```

ZIP Search Output

```
Zip Code: 56301, Place Name: Saint Cloud, State: MN, County: Stearns,  
Latitude: 45.555, Longitude: -94.167  
Zip Code 99999 not found in the file.
```

Troubleshooting

- **"Could not open file":** Check the file name and path (case-sensitive)
 - **Malformed data:** Ensure your CSV uses comma separators and proper quoting
 - **Missing header:** First row of CSV must match expected field names
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